

Unit 13, Lesson 11: What Is the Figure?



Benchmarks

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.



Learning Targets

- ☐ I can use coordinate geometry to classify triangles and quadrilaterals.
- ☐ I can use coordinate geometry to justify definitions, properties, or theorems of triangles and quadrilaterals.
- ☐ I can use coordinate geometry to solve mathematical problems involving triangles and quadrilaterals.



13.11.1 Warm-Up: Bell Work

Jonah is designing a stage in the shape of a trapezoid. Jonah has created two designs.

<i>VSZY</i>	<i>BGKR</i>
$V(2, -5)$, $S(4, -7)$, $Z(-2, -7)$, and $Y(-1, -5)$	$B(-4, -2)$, $G(-2, -2)$, $K(-7, -7)$, and $R(-7, -5)$

1. Determine whether Jonah can use either or both designs for the trapezoidal stage.
 - Show your work in an organized way.
 - Write your conclusion explaining the properties used to determine your answer.



13.11.2 Activity: Card Sort

Match each image, classification, coordinates, and property that represent the same polygon. Record your matches in the table by writing the letter or number of each card and showing your justifications.

Match 1	Match 2
Image:	Image:
Coordinates:	Coordinates:
Classification:	Classification:
Property:	Property:
Justification:	Justification:

Match 3	Match 4
Image:	Image:
Coordinates:	Coordinates:
Classification:	Classification:
Property:	Property:
Justification:	Justification:

Match 5	Match 6
Image:	Image:
Coordinates:	Coordinates:
Classification:	Classification:
Property:	Property:
Justification:	Justification:

Match 7	Match 8
Image:	Image:
Coordinates:	Coordinates:
Classification:	Classification:
Property:	Property:
Justification:	Justification:

Match 9	Match 10
Image:	Image:
Coordinates:	Coordinates:
Classification:	Classification:
Property:	Property:
Justification:	Justification:



13.11.3 Wrap-Up: Cool Down

1. Explain how you can use coordinate geometry to classify triangles and quadrilaterals.